

Challenges Faced In Designing 'APPROVED' MEDICAL DEVICES In India



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The medical device industry in India is fairly new. According to Infosys, around 75 per cent of the medical equipment is imported. And the equipment is not always suitable for Indian requirements.

Why are the medical devices not being made sufficiently in India? There are some unique challenges to be addressed here. Let us take a look.

Challenges involved in medical device design

The first problem that medical device designers encounter is that, they need to have interdisciplinary knowledge to be able to design good-quality products. It is not sufficient to have a good medical background, because every consumer medical product also has a substantial electronic component that needs to be designed. Thus, there needs to be a collaboration between the doctors and engineers to ensure that the product gives a good overall performance. This collaboration is challenging, because engineers and doctors are not usually

trained to work together.

The other problem is that, the medical device market is highly regulated. No hospital would purchase a medical product unless authorities approve it. Thus, the product has to meet all requirements of the regulatory authorities before it can enter the market.

We do not have a regulation set-up in India yet for medical regulation bodies. A bill proposed to solve this is due in the Parliament for approval. Central Drugs Standard Control Organization (CDSCO) is responsible for some level of device regulation in India.

There is also a new amendment in discussion with respect to existing rules, for which a draft has been introduced for public comments. Known as Medical Device Rules, 2016, it suggests categorisation of medical devices into four groups, depending on the risks associated with the device.

Dealing with the challenges

While designing a new medical device, it is important to do some research and consult with multiple doctors. This is called the process of concept validation. Information from doctors is very helpful. At this stage, they can provide feedback regarding clinical issues that the concept may have. The design can be improved to overcome these issues.

After the doctors approve the concept, the real design process begins, involving multiple levels of prototypes that have to be tested in various labs.

Documentation stage is very important for medical devices, and has to be done at every step of design control. One of these documents is the market requirement document. It includes all the results

Benefits of medical device innovation

One of the main advantages of designing a medical device is that returns on investment are high. Another advantage is that the market is still quite new and, so, chances of success are quite high, if the product is good. It is also a good opportunity to help society on a larger scale, with improved medical devices.

Good funding is available

Good funding is available for people interested in designing medical devices. Indian government is providing a lot of financial support in this area. Some grants are given to those having just a concept in mind, such as Biotech Ignition Grant, which is awarded by Department of Biotechnology. There are also certain private organisations and NGOs that are funding medical device companies such as Bill and Melinda Gates Foundation, Consortium for Affordable Medical Technologies and Acumen.